Liquid Crystals and the Technology they Power

Solid State Physics Final Project

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Outline

Orientation and Order Parameters

Phases and Phase Transitions

LCD Technology

Recent Developments in Liquid Crystals

Section 1

Orientation and Order Parameters

Phases and Phase Transitions

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Recent Developments in Liquid Crystals

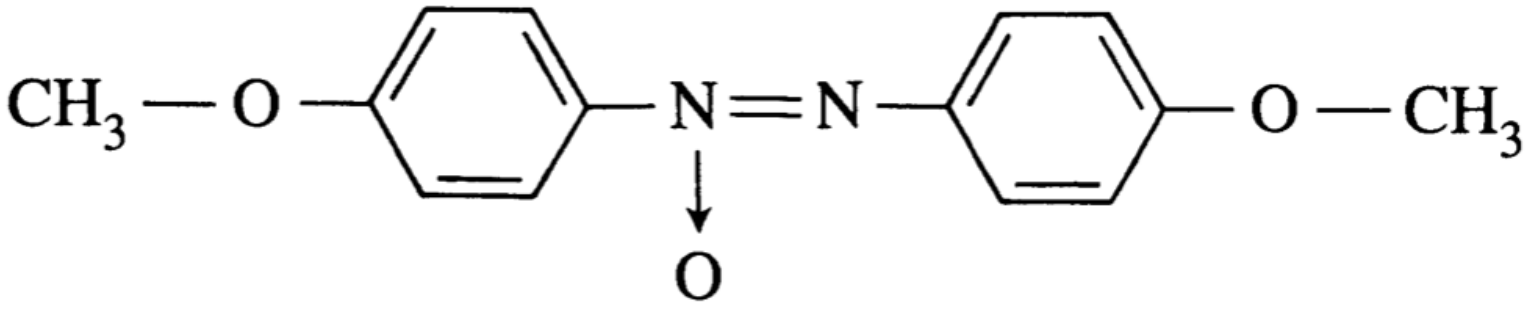
Liquid Crystals

Liquid crystals are a phase of matter that has orientation properties of a crystal, but is not confined to a lattice. They have short-range order, but not long-range order [2], and have two coherence lengths $\xi_{/\!/}$ and $\xi_{\perp}$ which are parallel and perpendicular to the major axis of the molecule.

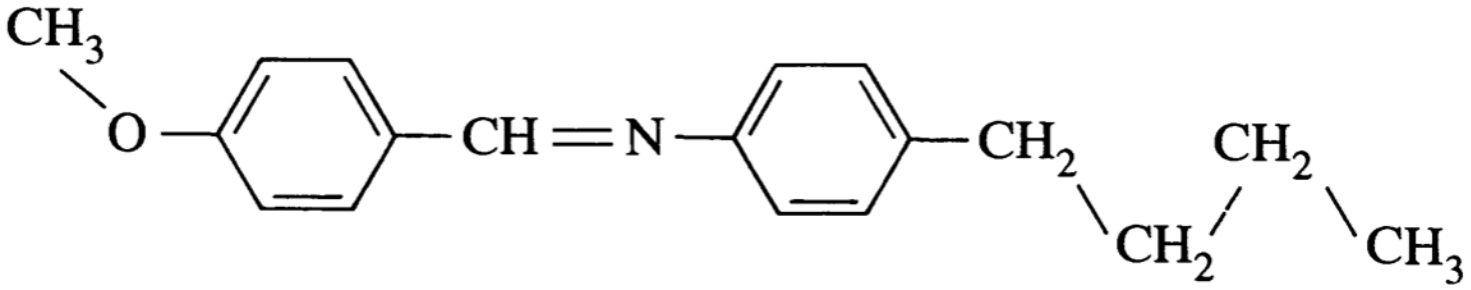
In this section, we want to motivate the concept of "average orientation" of molecules in a liquid crystal and discuss the distribution of orientations. In particular, we consider liquid crystals of rod-like molecules.

Molecular Shape

Liquid Crystals of rod-like molecules are usually made using cholesterol-like organic molecules. Here are a couple examples (DeGennes-Prost) [5]:



Above: p-azoxylanisole or PAA.
Below: N-(p-methyloxybenzylidene)-p-butylaniline or MMBA.



These give rise to nematic phases of liquid crystals- more later.

Molecule Orientation and Ergodicity

For a single molecule, define its pointing direction as $\hat{\ell}$. Here, we do not distinguish between $\hat{\ell}$ and $-\hat{\ell}$. We assume the molecule can move with respect to time, with average direction $\hat{n}$.

Now, we take an ensemble of molecules and assume they are *ergodic*, or that the time average of a single molecule is the spatial average of the ensemble. In symbols, this is

$$\frac{1}{T}\int_0^T \hat{\ell}(t)dt = \frac{1}{N}\sum_{k=1}^{N}\hat{\ell}_k = \hat{n},$$

where $\hat{\ell}_k$ is the instantaneous orientation for the kth molecule at time 0 and T is an arbitrarily large time.

Building up a Distribution

Our goal is to find a function $g(x, \Omega) = g_x(\Omega)$ such that for each $x \in \mathbb{R}^3$, $g_x(\Omega)d\Omega$ is the probability of the orientation at x being in the neighborhood $\Omega \pm d\Omega$, where Ω is a point on the unit sphere. This is usually called an *orientation distribution function* [7].

We can parametrize $\Omega = (\phi, \theta)$ or $\Omega = \hat{s} = (s_x, s_y, s_z)$ where $s_x^2 + s_y^2 + s_z^2 = 1$.

We can always write

$$g_x(\Omega) = \sum_{l,m} c_{x,lm} Y_{lm}(\Omega),$$

but we are going to make some assumptions that make our lives easier. First, we need $\int_{S^2} g_x(\Omega)d\Omega = 1$ in order for it to be a proper probability distribution.

Simplifying Assumptions

1. Since we identify $\hat{n} \leftrightarrow -\hat{n}$, we can only consider even-l spherical harmonics.
2. The distribution function should have azimuthal symmetry since in a bulk crystal, the deviation is rotationally isotropic. Thus, we only need to consider $l = 0$ and $l = 2$ spherical harmonics.
3. By the Spherical Harmonic Addition Theorem (below), we can change $g_x(\Omega)$ to $g_x(\cos(\theta))$, the dot product of Ω and $\hat{n}$. (Here, we assume $\hat{n}$ to be pointing in the $\hat{z}$ direction on the sphere.)

Spherical Harmonic Addition Theorem:

$$P_l(\hat{s} \cdot \hat{s}') \propto \sum_{m=-l}^{l} Y_{lm}(\hat{s}) Y^*_{lm}(\hat{s}').$$

Order Parameter!

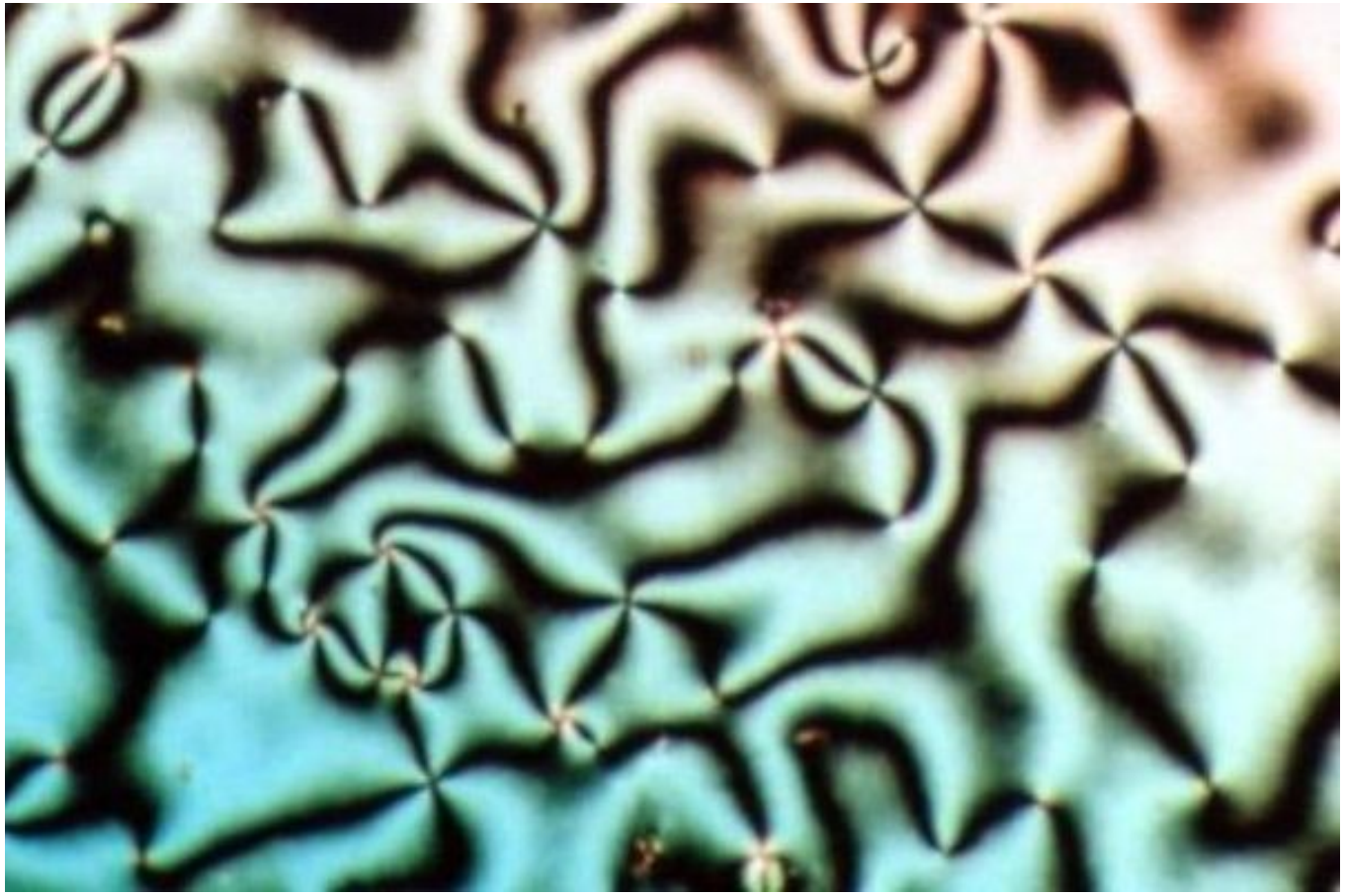
We've shown that the relevant quantity is the second spherical harmonic, which we can reduce to the second Legendre polynomial. Thus, while we can plot $g(x, \Omega)$ by plotting the distribution on a sphere for each x, we can reduce the number of data points and consider the dimensionless parameter

$$S_x := \langle P_2(\cos(\theta)) \rangle_x = \int_{S^2} g_x(\cos(\theta)) P_2(\cos(\theta)) d\Omega,$$

where $P_2(y) = \frac{1}{2}\left(3y^2 - 1\right)$. Here are some values for S_x for particular orientations.

Angular Distribution	$\left\langle \cos^2(\theta) \right\rangle$	S_x
Parallel alignment/Peaked around $\theta = 0, \pi$	1	1
Perp. alignment/Peaked around $\theta = \pi/2$	0	-1/2
Random alignment	1/3	0

So What?



This is a picture of a film of liquid crystals (Wikipedia). Changes in color indicate a change of average orientation, so the parameter S_x will change with color.
Also, we can directly measure S_x using NMR techniques!

Section 2

Orientation and Order Parameters

Phases and Phase Transitions

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Liquid Crystal Phases

Solid Phase: Not a liquid.

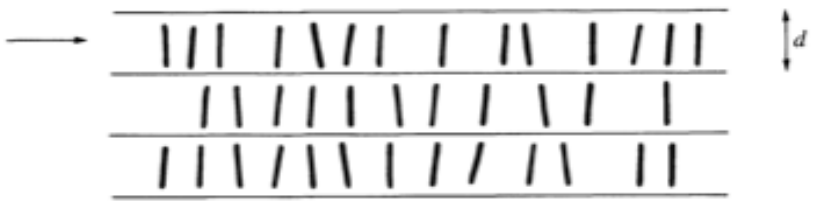
Isotropic Phase: Molecules are randomly arranged and randomly oriented.

Smectic Phase: Stacked layers of planar molecules; each layer is a two-dimensional liquid with random molecular arrangement and preferred alignment $\hat{n}$ within layers. If $\hat{n} /\!/ \hat{z}$, it is a smectic A phase, and if $\hat{n}$ is not parallel to $\hat{z}$, then it is a smectic C phase.

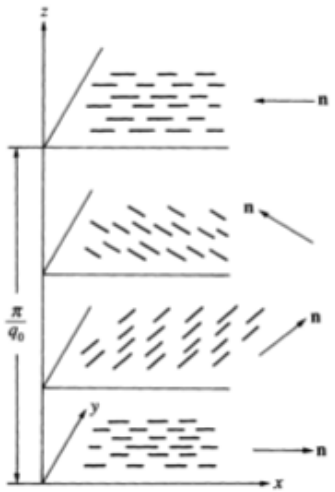
Nematic Phase: Centers of molecules are randomly arranged, and all molecules' pointing direction are distributed around $\hat{n}$.

Twisted Nematic Phase: At each z cross-section, the molecules look like a nematic phase with direction $\hat{n}(z) = \mathrm{Rot}^{(z)}_{\phi(z)}\hat{x}$.

Pictorial Representation of Phases



Smectic Phase



Twisted Nematic Phase

(Degennes-Prost)

State Variables I

Q: What variables should be used to describe a liquid crystal phase? Discuss amongst yourselves.

Examples: $P, V, T, S, Q, \ldots$

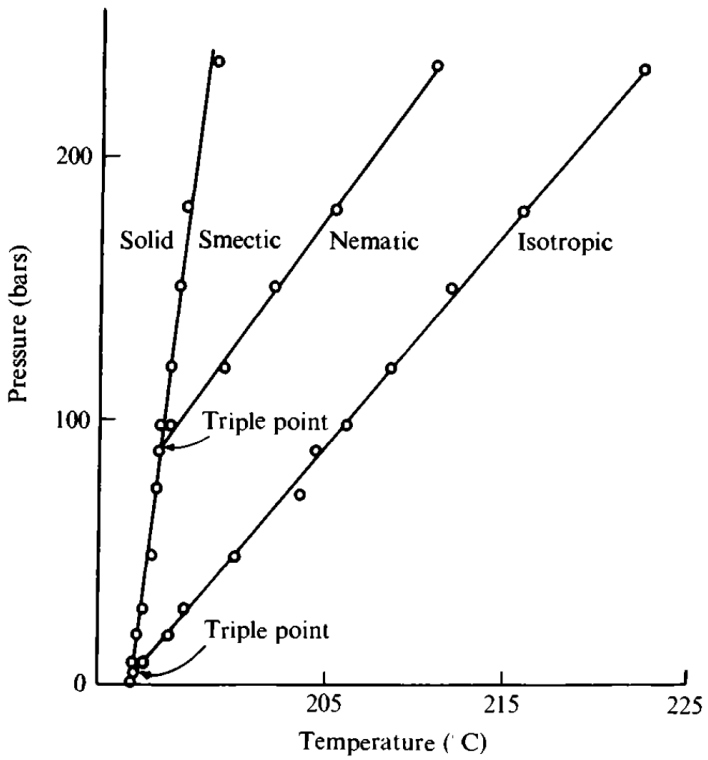
State Variables II

The main state variables we are considering are temperature and pressure.

Temperature: Needed to melt the solid into a liquid, and introduce thermal disorder in the system.

Pressure: If pressure is too low, there is no advantage to self-ordering. Isotropic state is preferred until fluid is put under pressure, where there is an energetic advantage to minimize effective volume.

Phase Diagrams and Triple Points



(Chandrasekar)

Triple point: a point (T, P) that borders more than 2 phases

You can think of a triple point as an unstable equilibrium point, where small perturbations in state variables lead to drastic physical changes.

Statistically Modelling Phase Transitions

In general, modelling these transitions is very complicated and requires nonlinear integral equations, even for low-order transitions. Here, I'll work out a nematic-smectic transition using a version of Ginzburg-Landau Theory, following Chandrasekar's exposition. We start with an expression for the smectic density wavefunction. [3]

$$\frac{\rho(z)}{\rho_0} = 1 + \frac{1}{\sqrt{2}} |\psi| \cos\left(\frac{2\pi z}{d} - \phi\right)$$

$\rho_0 =$ average density
$|\psi| =$ amplitude
$d =$ interlayer spacing
$\phi =$ phase factor

Complex Order Parameter and G-L Free Energy

From the last slide, we can encapsulate the order information in the complex parameter $\psi = |\psi| \, e^{i\phi}$.

Near the transition, we can expand

$$F_{LG} = \alpha\psi^2 + \beta\psi^4 \tag{1}$$

$$= a(T - T_c)\psi^2 + \beta\psi^4 + \frac{\delta S^2}{2\chi} - C\psi^2\delta S \tag{2}$$

Here, T_c is the critical temperature, $\beta > 0$, $C > 0$, S is the order parameter from Section 1, and χ is a response function that depends on saturation of nematic order.

The last two terms are added to couple ψ and S.

Minimizing Free Energy

Minimizing F_{LG} with respect to δS leads to the equation

$$F_{LG} = \alpha\psi^2 + \beta'\psi^4,$$

where $\beta' = \beta - \frac{\chi C^2}{2}$. If $\beta' > 0$, the transition is second-order. If $\beta' < 0$, the transition is first-order and highly unstable. (In this case, we need to expand to 6th order to ensure stability).

To find the triple point, we need $\beta' = 0$, which leads to $\chi(T_c) = \frac{2\beta}{C^2}$, which is consistent with microscopic theory.

Section 3

Orientation and Order Parameters

Phases and Phase Transitions

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Liquid Crystal Displays

They're everywhere! (Except for OLED displays - see LED talk/paper)

Electro-optical Properties

Liquid crystals are used for displays because their optical properties can be tightly controlled by the presence or absence of small electric fields and voltages; this is particularly salient for low power consumption displays.

In nematic crystals, the optical properties are determined by the pointing direction and the electric field.

In twisted nematic crystals, the optical properties are determined by the amount of twist in the crystal orientation.

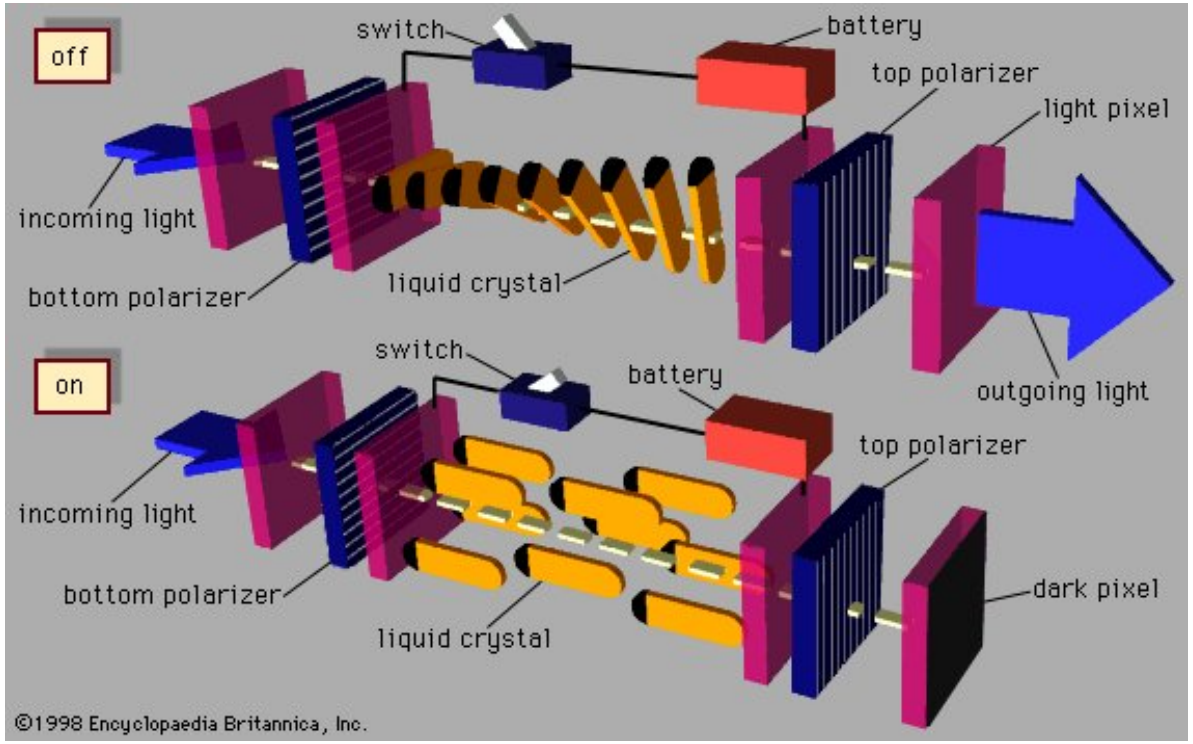
Nematic LCDs (1962: Richard Williams)

The most rudimentary technology for displays is with nematic crystals, where the optical properties are controlled by electric fields.

We apply an electric field to the crystals, and contrast is achieved by having crystals point parallel or perpendicular to the electric field. This requires an active current, so it was phased out in favor of more resilient technology. [8]

Twisted Nematic LCDs (1970-1971: Martin Schadt)

Twisted Nematic, or TN, LCDs are a robust technology that is being used even nowadays in displays that prioritize refresh rate over color depth. They use much less current than nematic LCDs since we only need small, local electric fields to power the display.



In the "off" state, the crystals are transparent to light, and in the "on" state, the crystals are opaque. [4] (Brittanica)

Electro-optical Properties II

Looking at the TN panel, one may wonder why the orientation of the crystals is related to having an on/off state. Since liquid crystal molecules have a dipole moment parallel to their pointing direction, we can make an analogy with polarized light from EM. (boardwork)

One can show that in EM, if we have two perpendicular polarizers back to back, no light can penetrate. However, if we have N polarizers that slowly rotate from vertical to horizontal, light can penetrate through.

In-Plane Switching LCDs (1992-1996: Katsumi Kondo)

In the 1990s, scientists at Hitachi and Samsung developed an improvement to LCD technology, where we have one overall substrate and use electrodes to locally untwist portions of the liquid crystal.

IPS panels usually have better color accuracy than TN panels (quantified by $\Delta E/E$ values), and require complex electrode circuitry to operate.

Section 4

Orientation and Order Parameters

Phases and Phase Transitions

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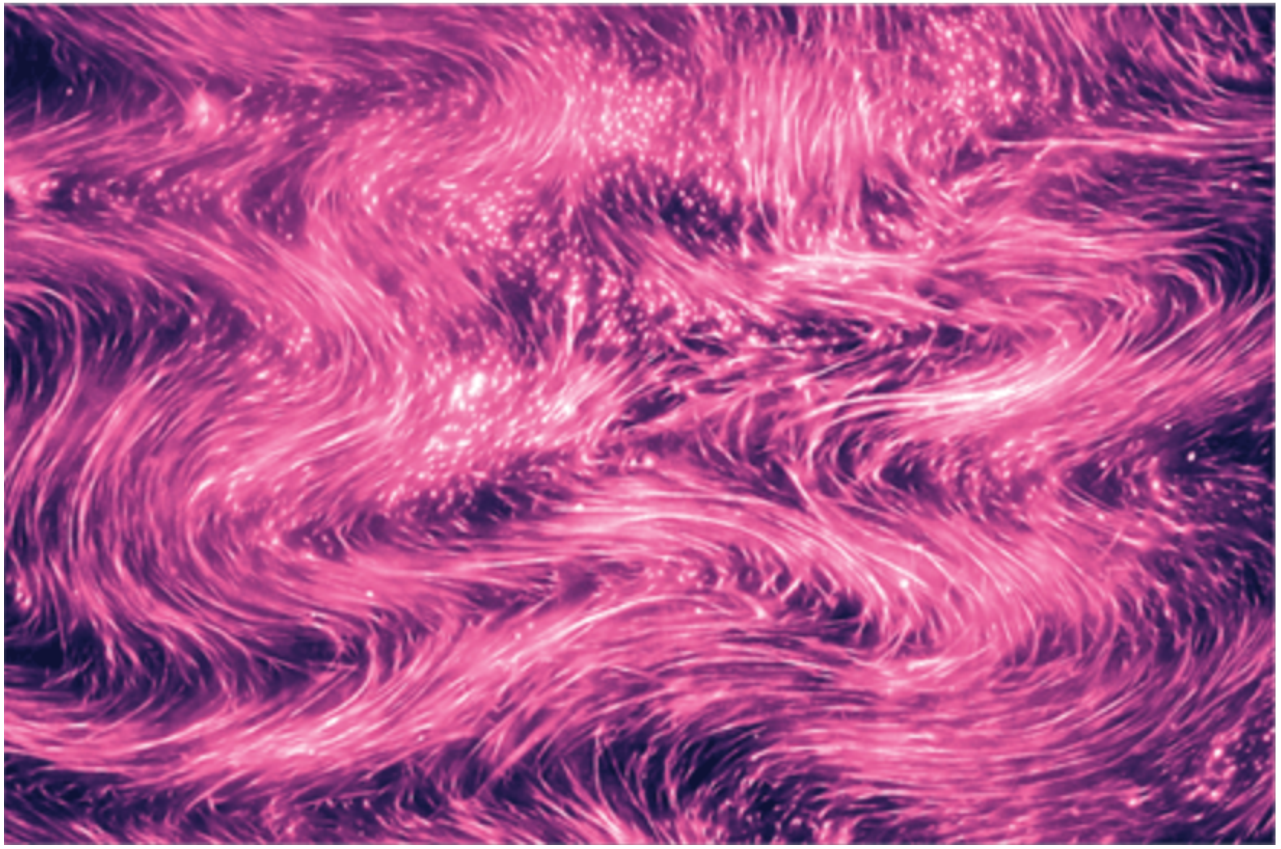
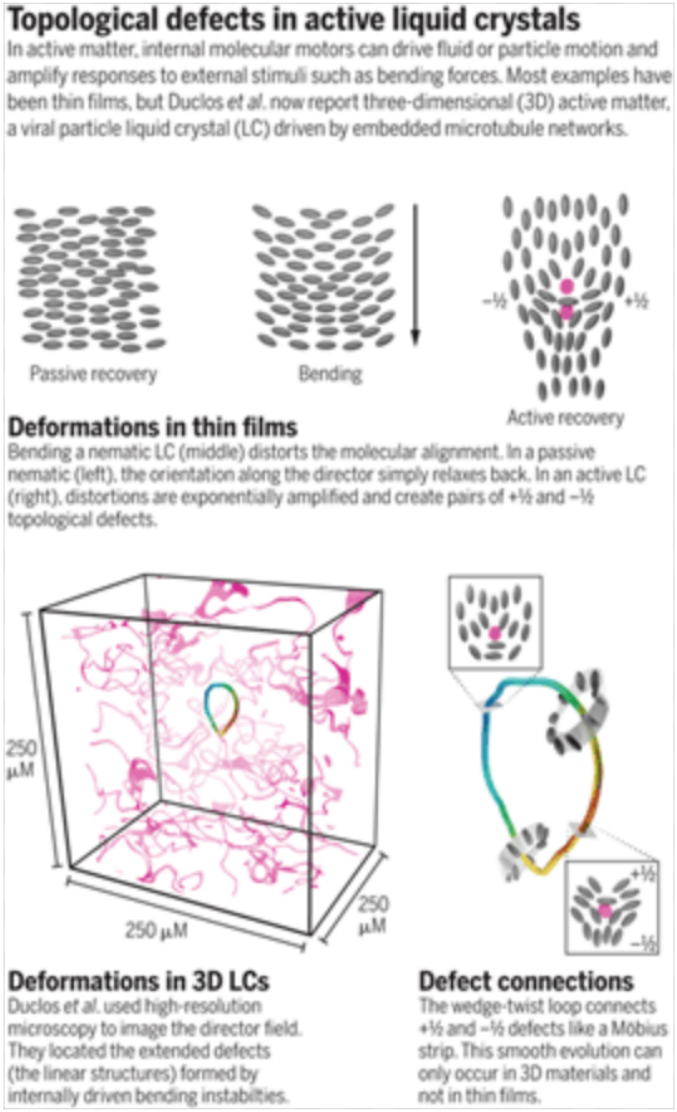
Liquid Crystals and Topology I



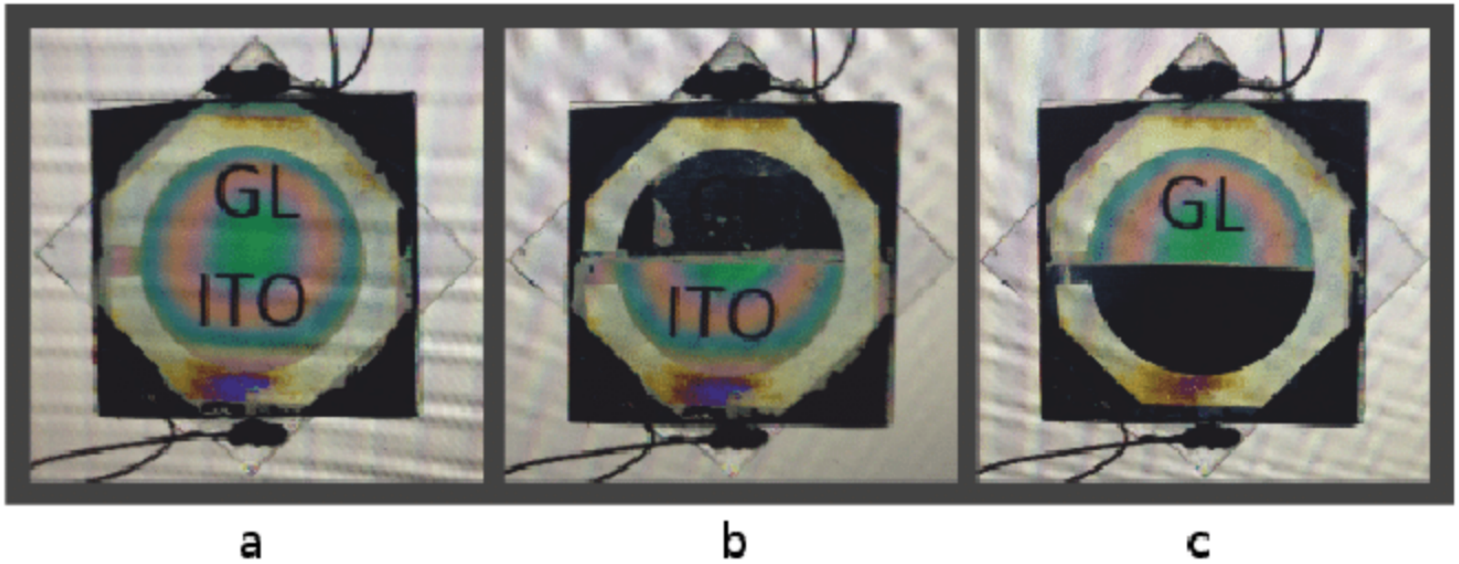
Groups are studying the fluid dynamics of nematic liquid crystals - the novel phenomenon is that there are no self-similar structures in the turbulence caused by shears and stresses. [1]

Liquid Crystals and Topology II



We can also analyze the vortex dynamics of liquid crystals, consider the homotopic properties of loops, and analyze the defects and phase boundaries of the crystal.

Graphene Electrodes



In order for LCDs to be flexible, the electrode technology has to change - here, a group show graphene electrodes as a more flexible alternative to standard Indium-Tin-Oxide (ITO) films. [6]

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